Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient's X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient's condition?

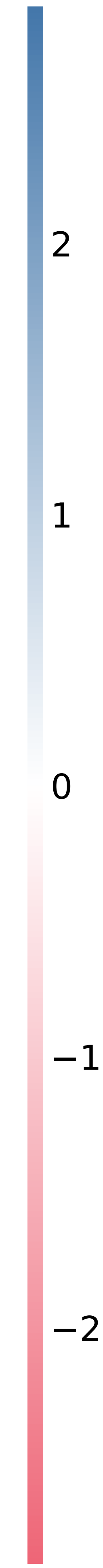
Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.



Reasoning + Answer:

<think> The patient's symptoms and the response to treatment suggest that the cause of the community-acquired pneumonia is a bacterial pathogen. The pathogen that is most likely causing this patient's condition is Streptococcus pneumoniae. The immune response elicited by Streptococcus pneumoniae is a classic example of a Th1-cell response. The Th1 response is a type of cell-mediated immunity that is involved in the defense against intracellular pathogens and is characterized by the activation of macrophages and the production of cytokines such as interferon-gamma. This type of response is important for the elimination of bacterial pathogens that enter the body through inhalation, such as Streptococcus pneumoniae. The other options are not the best description of the immune response elicited by the pathogen causing this patient's condition. Option A is correct for the immune response elicited by bacterial flagellin, which activates TLR5 on the surface of macrophages. Option C is correct for bacteria such as Staphylococcus aureus or E. coli, which have peptidoglycan as a major antigen that induces an immune response. Option D is correct for bacteria such as Streptococcus pneumoiae is not a example of pathogen that evades the immune system by encapsulation, that is more related to encapsulated bacteria like Streptococcus agalactiae (Group B streptococcus) or Streptococcus pneumoiae serotype 3 not the typical type </think> <answer> B. It primarily induces the Th1-cell response. </answer>